



Carpenters help in diagnosis and troubleshooting
(Credit: www.cbsnews.com)

System for Mobile Communications (GSM) or 3G.

Technologies for automotive electronic systems. These systems are used in vehicles for engine management, ignition, radio, computer-on-board (carputers), telematics, in-car entertainment, navigation and related elements of other control systems.

Automotive electronic systems could also include advanced driver-assistance systems like lane-assist, speed-assist, blind spot detection, park-assist, adaptive cruise control, pre-collision assist and so on.

Recent advances in hardware memory management and real-time operating systems have made it possible to use fewer and more capable computer microprocessors as a part of in-vehicle automotive electronic systems. A typical vehicle in the early 2000s had non-real-time microprocessor-based operating systems, whereas the new embedded system platforms include far more sophisticated networks and software applications, artificial intelligence (AI) and ubiquitous computing.

Technologies for traffic measurement. Various technologies are available to obtain raw data for traffic measurement. Data from different methods can be combined in intelligent ways to determine the state of the traffic accurately. A data-fusion-based approach that utilises road-side-collected acoustics, image and sensor data bring together the advantages of different individual methods.

Normally, cameras are used; however, today, modern, faster and more reliable methods using sensors are available for traffic measurement. These are less expensive than cameras, provide more coverage (potentially including all locations and streets), are faster to set up, need less maintenance and work

in all weather conditions, including heavy rain.

Video vehicle-detection system. Traffic-flow measurement and automatic incident detection using video cameras is the most popular form of vehicle detection. Cameras are mounted on poles or structures above or adjacent to roadways. This is a non-intrusive method of traffic detection, since it does not involve installing any components directly into the road surface or roadbed.

Video footage from the cameras is fed to processors, which analyse the changing characteristics of the video images as vehicles pass by. A single video-detection processor can detect traffic from one to eight cameras, simultaneously. The system requires some initial configuration to familiarise the processor with the baseline background image by feeding known measurements such as distance between lane lines or height of camera above the roadway as input. Output from a video vehicle-detection system includes lane-by-lane vehicle speeds, counts and lane-occupancy

readings. Some systems can also provide additional outputs such as gaps, headway, stopped-vehicle detection and wrong-way vehicle alarms.

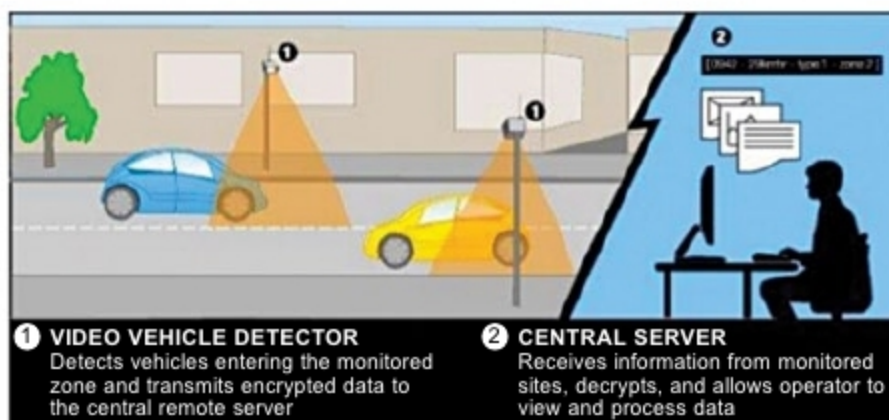
Audio vehicle-detection system. Traffic density can also be measured using the audio signal that consists of cumulative sounds from tyre noise, engine noise, engine-idling noise, honks and air turbulence noise. A roadside-installed microphone picks up the audio that comprises various vehicle noise, and audio-signal-processing techniques are applied to estimate the state of traffic. Accuracy of such a system compares well with other methods.

Sensing systems. Sensing systems are vehicle- and infrastructure-based networked systems. Infrastructure sensors are indestructible, such as in-road reflector devices that are installed or embedded in the road or at road sides. Vehicle-sensing systems deploy infrastructure-to-vehicle and vehicle-to-infrastructure electronic beacons for identification communications. These may also employ video automatic number-plate-recognition or vehicle magnetic-signature-detection technologies at intervals to increase sustained monitoring of vehicles in critical zones.

Inductive loops are placed in a roadbed to detect vehicles as these pass through the loop's magnetic field. The simplest detectors simply count the number of vehicles during a unit of time that pass over the loop, while more sophisticated sensors estimate the speed, length and class of vehicles, and the distance between these. Loops can be placed in a single lane or across multiple lanes. These work with very slow or stopped vehicles, as well as vehicles moving at high speed.

Systems based on cell-phones. These days a high proportion of vehicles contain one or more cell-phones that periodically transmit their presence information to the cell-phone networks, even when no voice connection is established. As a vehicle moves, so does the signal

Video from cameras is fed into processors that analyse the changing characteristics of the video image as vehicles pass. The cameras are typically mounted on poles or structures above or adjacent to the roadway.



Video vehicle-detection system (Credit: www.slideshare.net)